Alessandro Riina

Digital Humanities · Serious game design and educational evaluation

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PROFILE

I design digital experiences for education and cultural heritage — and I evaluate whether they work. I combine a background in cultural heritage with game design, data analysis and natural language processing. My most recent work is Custodi del Flascio, a serious game on biodiversity built during my internship at ARPA Sicilia, the regional environmental protection agency: it is live, it is the subject of my master's thesis, and it went through an expert evaluation with three primary-school teachers.

EXPERIENCE

Serious Game Designer & Developer — ARPA Sicilia, Palermo

University internship · January – April 2026 · 300 hours

- Designed and developed serious games on Sicilian biodiversity for the regional environmental protection agency.
- Ran a state-of-the-art review of ecological serious games and scouted existing solutions in the field.
- Gathered and structured data on the habitats of Nebrodi Park, Bosco del Flascio and Plemmirio, aligning proposals with the agency's outreach goals.
- Wrote the Game Design Document and evaluated development environments, building playable prototypes in Twine and GDevelop.
- Ran internal playtests, analysed results and balanced the game mechanics.
- Implemented in-game telemetry for quantitative analysis of user interaction.
- Produced the final technical documentation and project bibliography.

Staff writer and graphic designer — iCrewPlay, Palermo

March – December 2021

- News, columns, feature articles and guides on video games.
- Graphic production supporting the editorial team.

Museum reception and invigilation — Fondazione Sant'Elia, Palermo

Internship · July 2021 · Palazzo Sant'Elia

- Visitor flow management and monitoring of exhibition rooms.
- Operation of the multimedia equipment supporting the exhibitions; reception of institutional guests.

Live reporter — The Shield of Wrestling

December 2020 – June 2021

- 94 articles in seven months: live reports written and published as events aired, on a fixed weekly schedule.

OTHER EXPERIENCE

Assistant store manager — La Toscana IJS, Netherlands

July – October 2023 · Schijndel, Son en Breugel, Best

- Staff coordination and day-to-day management of three shops.

PROJECTS

Custodi del Flascio · custodi-del-flascio.vercel.app

A management serious game on the biodiversity of Natura 2000 site ITA070007 "Bosco del Flascio", for the last two years of primary school. Content, species and pressures are drawn from the site's own conservation documentation under the EU Habitats Directive 92/43/EEC. React, Next.js, TypeScript, Tailwind CSS; source released under the MIT licence.

iMuseiDiPalermo · imuseidipalermo.wordpress.com

A public guide to the museums of Palermo: a seven-family thematic taxonomy, detailed entries, an events section and internal search. Built on WordPress.

Italian cultural heritage in numbers

Analysis of Italian Ministry of Culture open data (1996–2018): integration of four sources, descriptive statistics, Pearson correlation, inferential testing and time-series analysis with a moving average. Python, pandas, seaborn, scipy.

Classifying newspaper voice

Linguistic analysis with spaCy and text classification with a language model: few-shot against zero-shot prompting, label-order bias testing, a random baseline, confusion matrix and error analysis. Hugging Face, Gemma, scikit-learn.

EDUCATION

MA in Digital Humanities for the Cultural Industry (LM-43)

University of Palermo · 2026 · 110/110 cum laude

Thesis: “From game design to educational evaluation: Custodi del Flascio as a serious game for environmental education” — supervisor prof. Giuseppe Città.

BA in Cultural Heritage: Knowledge, Management, Enhancement (L-1)

University of Palermo · 2024 · 88/110

Final paper: “Museums and cultural venues: augmented reality supporting children, and its economic impact”.

Classical secondary school diploma

Istituto Don Bosco, Palermo · 2017

SKILLS

Game design and learning — serious game design with the LM-GM framework; Game Design Documents; mechanics balancing; analogue game design; Twine, GDevelop.

Research and evaluation — ex-ante expert evaluation; semi-structured interviews; deductive and inductive coding schemes; deviant-case analysis; playtesting and results analysis; Zotero.

Data and language — Python (pandas, seaborn, matplotlib, scipy); NLP with spaCy, Hugging Face and NLTK; LLM prompting and evaluation; cultural open data.

Development and web — React, Next.js, TypeScript, Tailwind CSS; HTML and CSS; Git, GitHub, Vercel, GitHub Pages; WordPress; responsive design.

Publishing and content — book series design and editorial workflows; writing under deadline; Business Model Canvas; GIMP, Canva, Photoshop.

LANGUAGES

Italian native · English B2 (CLA certification, University of Palermo, 2024)

CERTIFICATIONS

Python Pro – Complete Guide (Udemy, 2025) · Digital Skills: Artificial Intelligence (Accenture Future Learn, 2025) · Fundamentals of Digital Marketing (Google, 2025)